

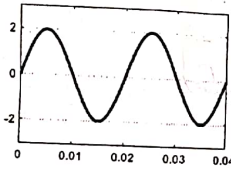
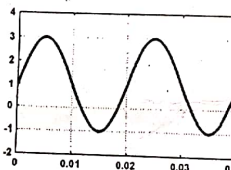
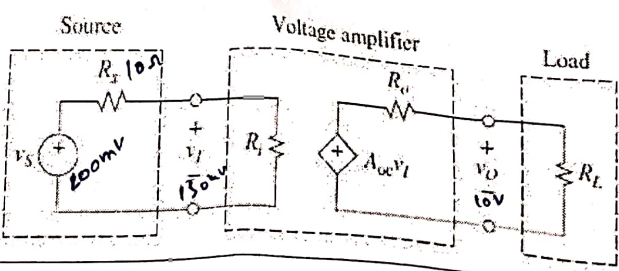
MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
Department of Electrical Engineering

B. Tech. IV Semester Main Examination May -2025
Sub: Electronics Devices & Circuits (EE-223)

Max. Marks: 50

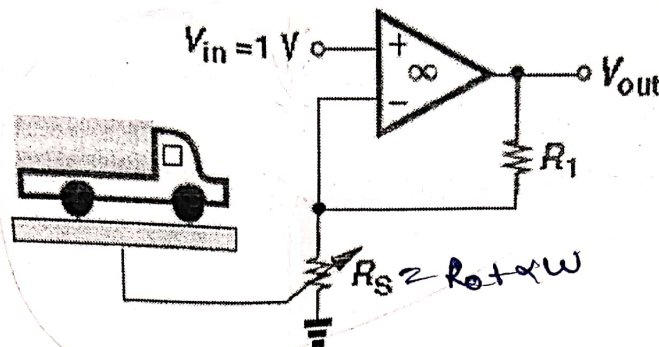
Note: Solve all questions else do as directed. Assume any missing data.

Time: 03 Hours

Q.1	Choose any five parts of the question, each part carries 02 marks.	CO5	
(i)	Tabulate the OP-AMP 741 parameters for: a. Open-loop voltage gain b. Input impedance c. Lowest operating frequency d. Highest operating frequency e. Slew rate f. Output impedance i. Common-mode voltage gain	CO1	02x5
(ii)	Draw output waveform of an Op-Amp oscillator having Wein bridge with $R = \left(\frac{7}{22}\right)k$, and $C = 1\mu F$ for Amplifier's gain (i) 2 and (ii) 3.5. The OP AMP supply voltage are $V_{cc} = +12V$ and $V_{ee} = -12V$.	CO2	
(iii)	In feedback topologies, the amplifier and the feedback-network respectively has gain of 2000 and 0.1. Estimate the factor by which input resistance R_{in} , the output resistance R_{out} and Bandwidth BW shall be modified in Voltage series feedback topology.	CO2	
(iv)	Implement an OP AMP circuit that produces output waveform from the input waveform shown below.	CO2	
  Input Waveform Output Waveform			
(v)	In a PCB board, there are 25 D flip flops used to assemble a system. If the V_{cc} of the Flip flop is 15V and the I_{cc} is 25mA, then what rating of DC power supply will be required to drive this system?	CO2	
(vi)	Multiply 01101010 by 11110001 in the 2's complement form.	CO1	
(vii)	Use a Karnaugh map to reduce each expression to a minimum SOP form: $\overline{AB}(\overline{CD} + \overline{CD}) + AB(\overline{CD} + \overline{CD}) + \overline{AB}\overline{CD}$	CO2	
Q.2(a)	Give the block diagram of 555 timer and discuss its working as Astable and Monostable Multivibrator. OR Discuss the working principle of OP-Amp based Astable and Mono-stable Multivibrator.	CO3	05
(b)	A voltage amplifier of the type of Fig. below is fed by a 200-mV source and drives a 10Ω load. Voltmeter measurements give $v_i = 150mV$ and $v_o = 10V$. If disconnecting the load causes v_o to rise to 12V, and then connecting a $30-k\Omega$ resistor across the input-port terminals causes v_o to drop from 12V to 9.6V, find R_s , R_i , A_{oc} , and R_o .	CO2	05
			

Q3(a)

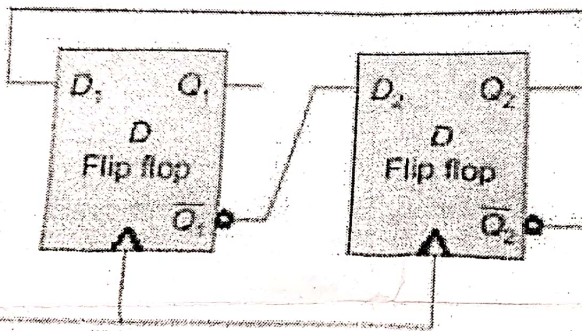
The truck weighing-station in the Fig. below incorporates a sensor whose resistance varies as $R_s = R_0 + \alpha \cdot W$, where R_0 is a constant value, α is a proportionality constant & W the weight of the truck. If $V_{in} = 1$ V, determine the expression of gain, if gain of the system is defined as $\frac{dV_{out}}{dW}$.



CO3

Q4(a)

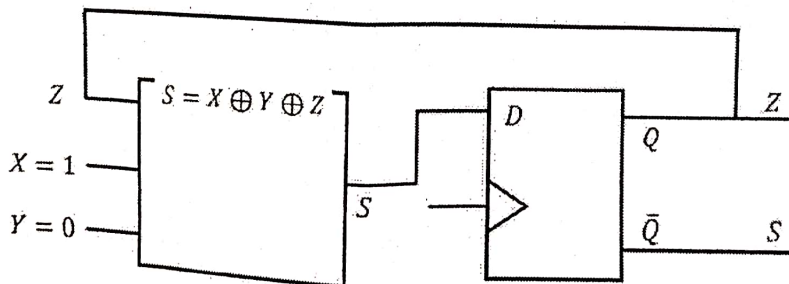
Design a digital hardware circuit to implement the line of code "count = count + 1" written in the frontend of the system



CLK = 1 MHz

In the given sequential circuit, the initial states are $Q_1=1$ and $Q_2=0$. For a clock frequency of 1MHz, draw the output waveforms Q_1 and Q_2 . Also determine the frequency of Q_1 and Q_2

(i) In the circuit, the present value of Z is 1. Determine the values of S and Z , respectively, after the application of the clock. Draw necessary waveforms

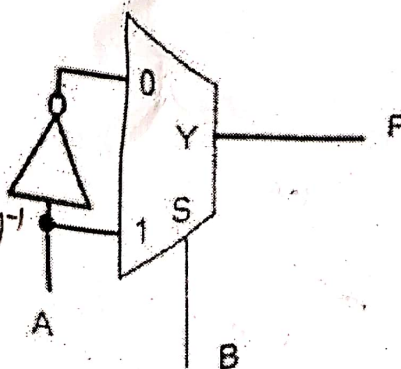


(ii) Consider the following circuit which uses a 2-to-1 multiplexer as shown in the figure below. What is the Boolean expression for output F .

$$\frac{dV_{out}}{dW} = \frac{d}{dW} \left(1 + \frac{R_1}{R_s} \right)$$

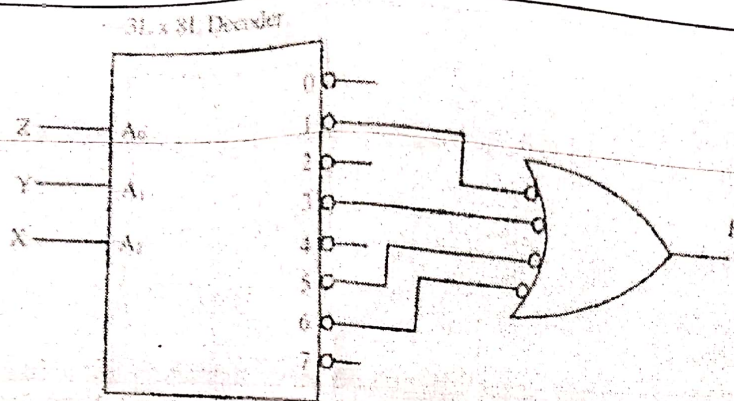
$$\frac{d}{dW} = 1 + R_1 \cdot \frac{d}{dW} \left(\frac{1}{R_0 + \alpha W} \right)$$

$$\frac{d}{dW} = 1 - \frac{R_1 \alpha}{(R_0 + \alpha W)^2}$$



A_0	A_1	S
0	0	0
0	1	1
1	0	0
1	1	1

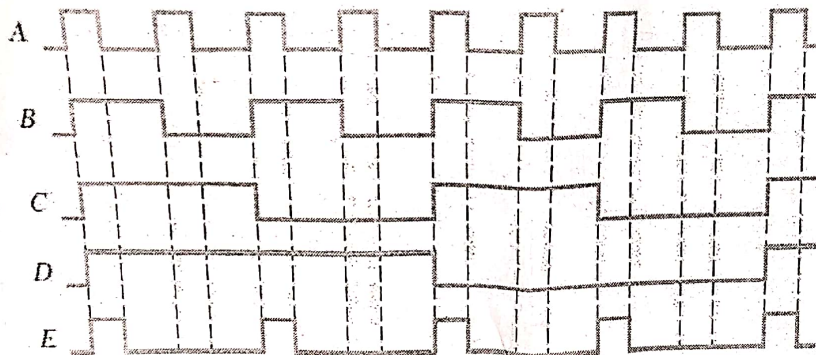
Q.5(a)



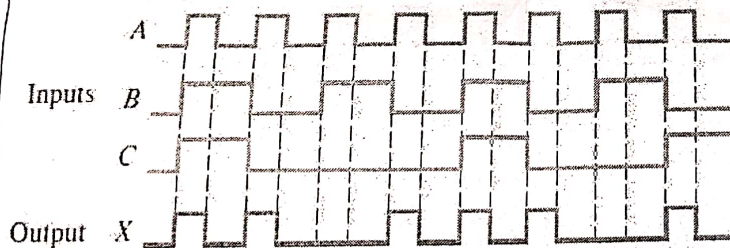
A 3 line to 8 line decoder, with active low outputs, is used to implement a 3-variable Boolean function as shown in the figure. What is the simplified form of Boolean function $F(A,B,C)$ implemented in "Sum of Product" and 'Product of Sum' form. Draw necessary waveforms

Q.5(b)

(i) For the waveforms given, A and B are ANDed to produce output F, D and E are XORed to produce output G. Then C, F, and G are ORed to produce output H. Draw the output waveform F, G and H.



(ii) Design a logic circuit that will generate the output waveform shown for the input waveforms shown in Figure below



Course Coordinator: Dr. T. S. Bheemraj
Date of Examination: 15.05.2025